

JARED LEMLER

Backend Engineer • Pacifica, CA

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PROFESSIONAL SUMMARY

Backend Engineer with a craftsman's approach and 8 years in electrical systems before software. My electrician background honed my systems thinking by coordinating teams, reading complex plans, and building infrastructure that can't fail, and I bring that same patience and troubleshooting discipline from wiring buildings to designing APIs, databases, and containerized services. Technically grounded in Python/FastAPI and TypeScript/Fastify, with production experience in PostgreSQL, Prisma, and Docker. Recently architected VoxPort, a federated podcast platform on the AT Protocol, building high-throughput data pipelines (1,000-event bounded in-memory queue, 100-event batched persistence, ~95% I/O reduction) under real-world constraints.

FEATURED PROJECTS

VoxPort (voxport.live) — Federated Podcast Platform • Fastify, Next.js, ATProto

- **Zero-Downtime Federation:** Built a resilient WebSocket consumer surviving 10 reconnection cycles with exponential backoff and cooloff, featuring a bounded memory backpressure queue (1,000-event cap per worker) processing 6 collections with 100-event batching to achieve a ~95% I/O reduction vs. naive per-event persistence.
- **Engagement Analytics Engine:** Designed and implemented an hourly velocity aggregation system tracking user engagement across 1h to 168h (7-day) windows using a custom weighted scoring algorithm (likes=1x, mentions=5x) for real-time trending content detection via scheduled background tasks.
- **Memory-Optimized Cursor Tracking:** Eliminated per-event database write bottlenecks by introducing an in-memory cursor system with 5-second flush intervals, successfully reducing database round trips by ~95% during high-volume ingestion spikes.
- **Relational Data Architecture:** Designed a fully normalized 26-entity database schema complete with dedicated time-series aggregation tables specifically optimized for historical trend analysis and real-time performance queries.
- **Scalable Media Infrastructure:** Integrated Cloudflare R2 for content-addressed blob storage, seamlessly mapping Content Identifiers (CIDs) to AT Protocol records for verifiable, decentralized media distribution across federated network nodes.
- **Distributed Extraction Worker:** Built a standalone Fly.io background worker claiming MediaJob records via FOR UPDATE SKIP LOCKED, featuring per-source-host politeness gating (1 concurrent job per host), exponential backoff retry policies, and graceful shutdown handling to keep heavy ffmpeg processing isolated from the API layer.
- **Zero-Trust Audio Security:** Implemented magic-number file validation, automated ffmpeg metadata stripping, and 450MB upload limits to block spoofed payload uploads while strictly preserving creator privacy.

doc-lok (github.com/JL037/doc-lok) — Pre-Prompt Context Condenser • TypeScript, Node.js, Zero-Dep

- **Context Window Optimization:** Engineered a TypeScript pre-prompt middleware library that condenses prompt payloads by resolving remote Markdown links into compact cryptographic markers (<lok_id>), significantly cutting context token consumption and downstream API latency.
- **Conditional Caching Pipeline:** Constructed a two-phase network verification engine using HTTP HEAD probes and If-None-Match ETag validation to completely bypass full payload downloads for unchanged remote documentation resources.
- **Zero-Dependency Architecture:** Authored a lightweight CLI tool and standalone npm library relying exclusively on native Node.js APIs, completely eliminating third-party supply-chain vulnerabilities and minimizing cold-start overhead. Published on npm.

Travel Wishlist (travelwishlist.app) — Location-Based Travel Planner • FastAPI, React, Docker, Postgres

- **Architecture & Design:** Built a full-stack application leveraging FastAPI, PostgreSQL, and Docker. Designed robust JWT authentication workflows, Pydantic data validation schemas, and a reusable service-layer architecture for CRUD operations.
- **External API Integrations:** Built a city-based weather lookup system integrating external weather APIs delivering ~100ms response times alongside interactive mapping layers for geographic destination visualization.

- **Calendar Itinerary System:** Engineered monthly and agenda calendar views allowing date-based trip organization, destination bookmarking, and scheduled travel itineraries.

PROFESSIONAL EXPERIENCE

Backend Developer (Self-Employed) — Talk

Jan 2025 – Feb 2026

Social Connection Platform • Netherlands / Pacifica, CA

- Developed and deployed a privacy-compliant web platform supporting 200+ active university students using React, Vite, Tailwind CSS, and a custom FastAPI backend infrastructure.
- Translated founder vision into rapid technical implementations by converting interactive Lovable prototypes into production-ready full-stack software solutions.
- Engineered a custom QR-code trigger bridging physical events with digital experiences for social experiments, enabling real-time user participation and automated data collection.
- Constructed automated data processing pipelines and visual analytics dashboards converting raw event metrics into actionable business insights, reducing manual reporting effort by 90%.
- Consulted directly with product team members and beta testers to optimize user onboarding workflows, directly increasing user retention during live trial runs.
- Collaborated closely with co-founders to align technical analytics and engagement telemetry with broader research goals and product strategies.

Technologies: Python, FastAPI, TypeScript, React, Tailwind CSS, Vite, Cloudflare, Plausible Analytics

Journeyman Electrician — Gaylor Electric • Wolverine Electric • IBEW

2017 – 2025

Indiana & Michigan

- Coordinated site teams and crews of 2–4 electricians on high-visibility commercial jobsites, reading and interpreting complex electrical and architectural blueprints to deliver projects on schedule and strictly to code.
- Built, maintained, and inspected critical electrical infrastructure under zero-tolerance failure standards, developing deep live-system troubleshooting capabilities under heavy time constraints and strict safety environments.
- Completed comprehensive trade school coursework through Associated Builders and Contractors (ABC) to earn official Journeyman status; remain an active dues-paying IBEW union member eligible for dispatch.

SKILLS & TECHNICAL COMPETENCIES

Languages & Core Databases: Python, JavaScript, TypeScript, SQL | PostgreSQL, Prisma Postgres, Neon, Alembic, Schema Data Modeling, Index Optimization

Frameworks & Cloud Infrastructure: FastAPI, Fastify, SQLAlchemy 2.0, Pydantic, Django, React, Prisma ORM | AWS (EC2, Elastic Beanstalk, S3), Cloudflare R2, Fly.io, Hetzner

DevOps, Data Pipelines & Concepts: Docker, Git, GitHub, JWT, pytest, Sentry, OpenTelemetry, Axiom | Pandas, ETL Pipelines, Batch Processing | REST API Design, Distributed Systems, AT Protocol, WebSockets

EDUCATION & CERTIFICATIONS

- **Journeyman Electrician Certification:** Associated Builders and Contractors (ABC) Indiana • Completed multi-year vocational trade training combining rigorous technical classroom instruction with on-the-job industrial experience to achieve full Journeyman status. Active dues-paying IBEW member.
- **Self-Directed Software Engineering Discipline:** Advanced ongoing practical study of Python, FastAPI, Fastify, Next.js, TypeScript, PostgreSQL, distributed systems, the AT Protocol, and beginning Rust. Built through shipping live production software systems (VoxPort, doc-lok, Travel Wishlist) and 1:1 mentorship from senior engineers including former Facebook and Microsoft engineers and a current Prisma platform architect.

WRITING & COMMUNITY

Published an AT Protocol developer skill ("atproto-pro") on skills.sh; write on pckt.blog about building on the AT Protocol, including a deep dive on AT Protocol & RSS feed integration.